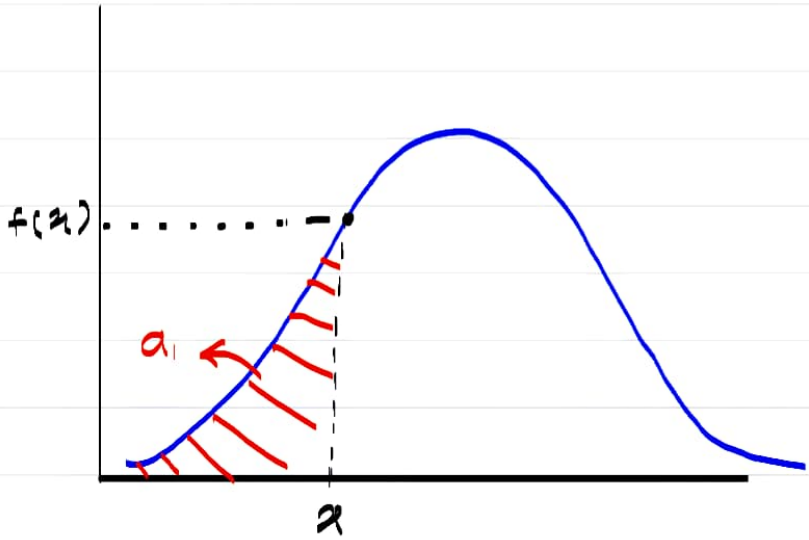
Advanced statistics

27-01-26



$$PDF = f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{2\sigma^2}\right)^{1/2}}$$

CDF = area under the curve

$CDF(x) = a_1$

Probability of $X \leq x$ = proportion of data between $-\infty$ to x

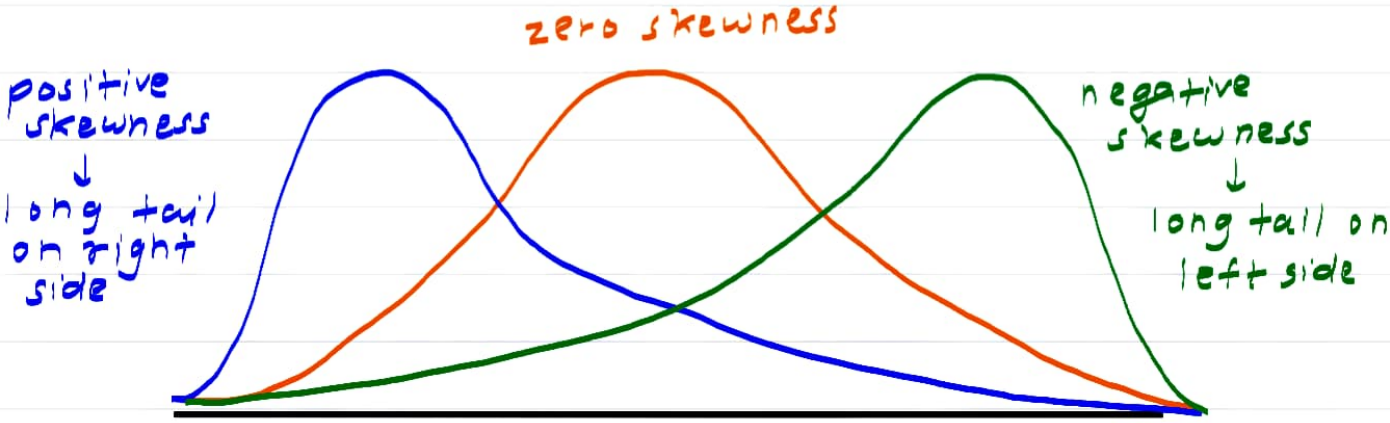
$$CDF(X=x) = \int_{-\infty}^{x} PDF(x)$$

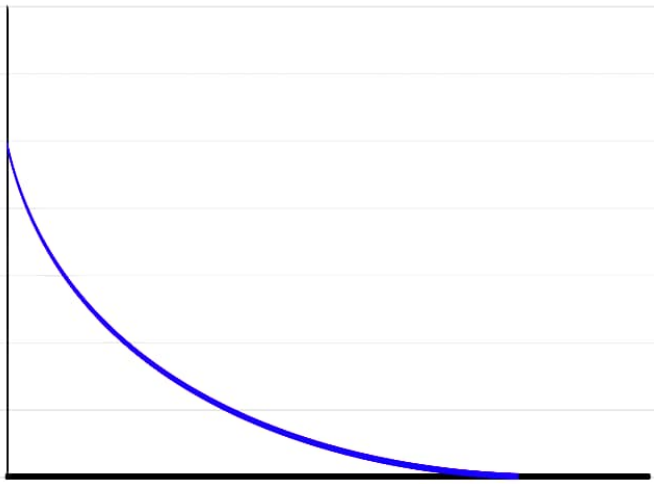
Skewness and Kurtosis

Skewness

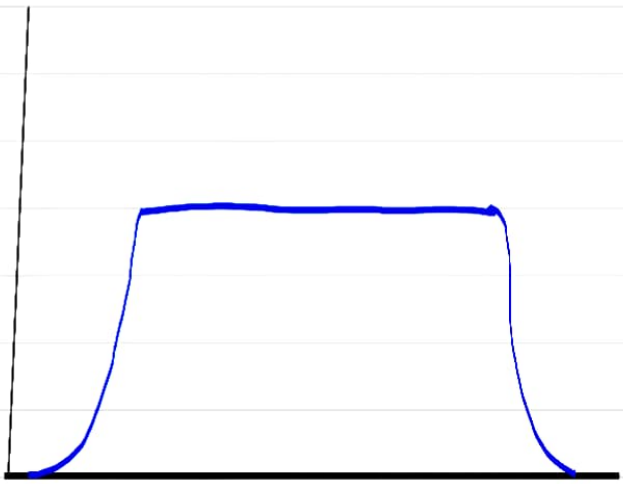
- it measures assymetry of data distribution and tells whether dist. is symmetric, left/right skewed.

negative skewness - dist is left skewed
positive skewness - dist. is right skewed
zero skewness - symmetrical dist. / normal dist.





exponential dist - positive skewness or right skewed



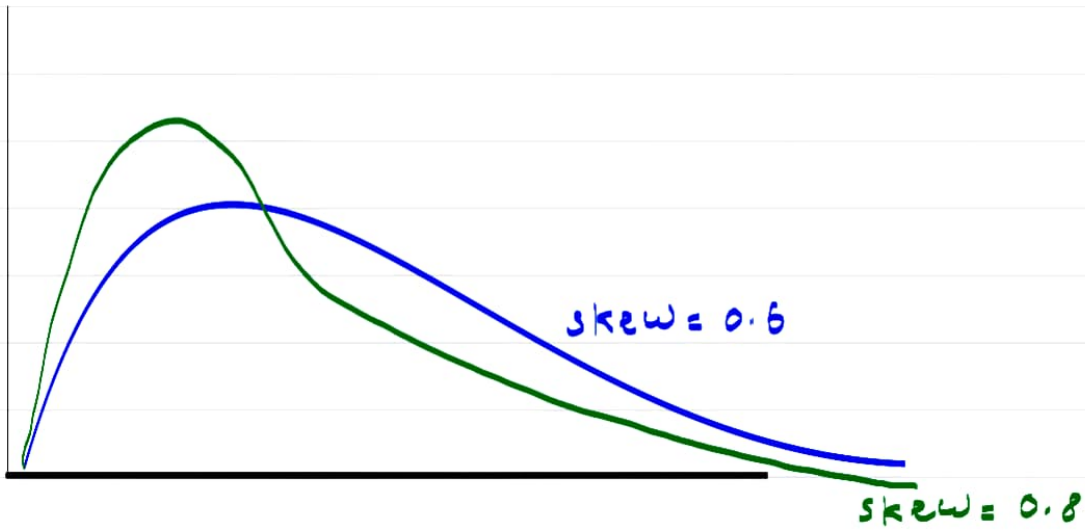
uniform dist - zero skewness or symmetrical

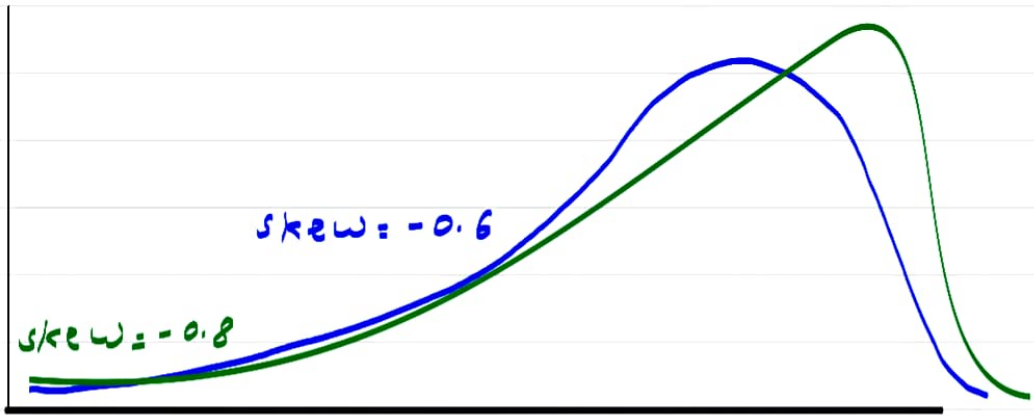
eg.

skewness values:

0.8 - right skewed
0.7 - right
-0.3 - left

0.0001 - symmetrical as skew ≈ 0
-0.0002 - symmetrical as skew ≈ 0





Kurtosis

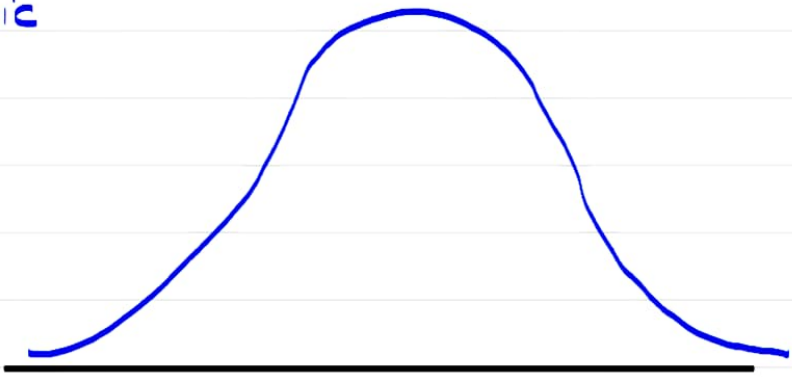
- describes shape of head and base of data distributn

- There are 3 types of shapes of data dist. based on kurtosis

1) mesokurtic

- kurtosis value is close to zero
- shape is very similar to a normal dist.
- it has moderate height of head and moderate width of base.
- in some frameworks kurtosis ≈ 3 is mesokurtic.

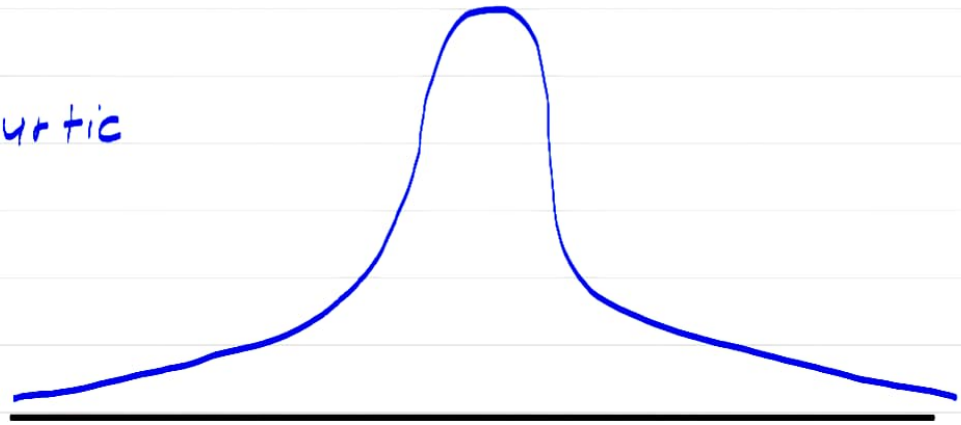
mesokurtic



2) <u>Leptokurtic</u>

- kurtosis value is greater than 0
- in some frameworks kurtosis > 3 is leptokurtic.
- leptokurtic dist has heavy tails and taller head.

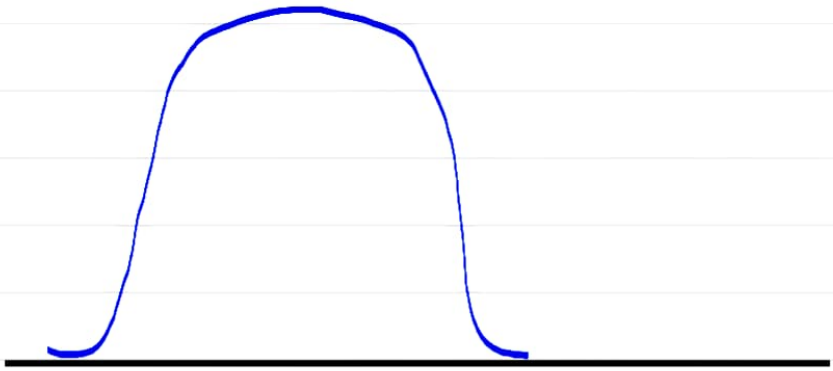
leptokurtic



3) <u>platykurtic</u>

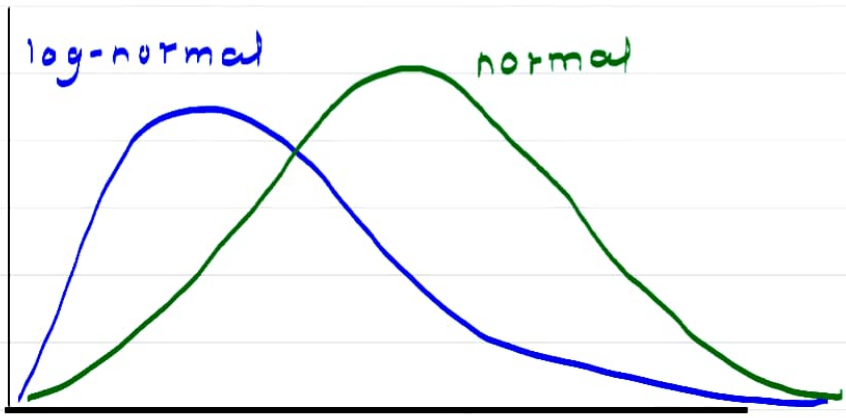
- kurtosis value is less than 0
- in some frameworks kurtosis < 3 is platykurtic
- platykurtic has very thin and light tails and head is short

platykurtic



Log-normal distribution

- its a continuous dist.
- its highly right skewed
- entity which can only have positive values can follow log normal dist.



log function

$a^x = y$ then $\log_a(y) = x$ what power of $a = y$

$3^2 = 9$ $\log_3(9) = 2$

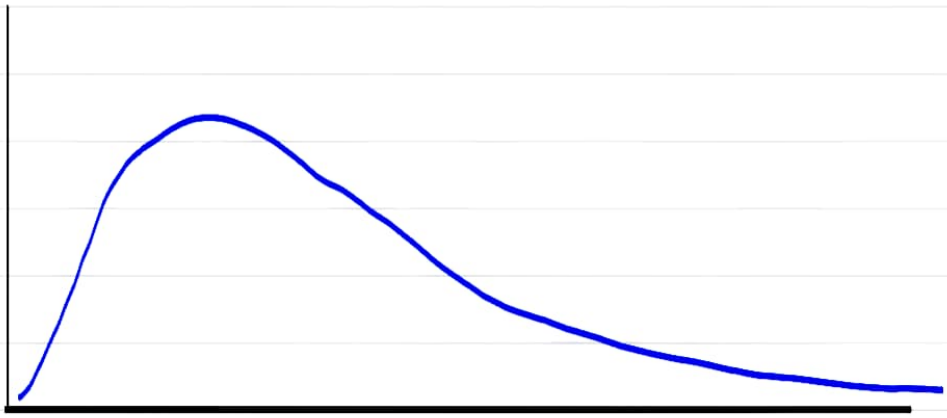
$2^4 = 16$ $\log_2(16) = 4$

$4^3 = 64$ $\log_4(64) = 3$

$\log_5(125) = 3$

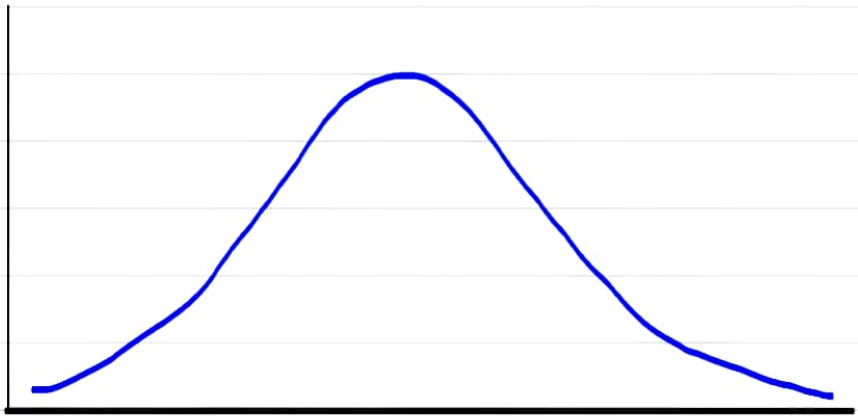
$\log_4(16) = 2$

data - [0.1, 0.12, 1.34, 4.56, 10.2, 100.12, 105.6 . . .]



log data = [$\log_e(0.1)$, $\log_e(0.12)$, $\log_e(1.34)$, $\log_e(4.56)$ . . .]

= [a_1 a_2 a_3 . . . . .]

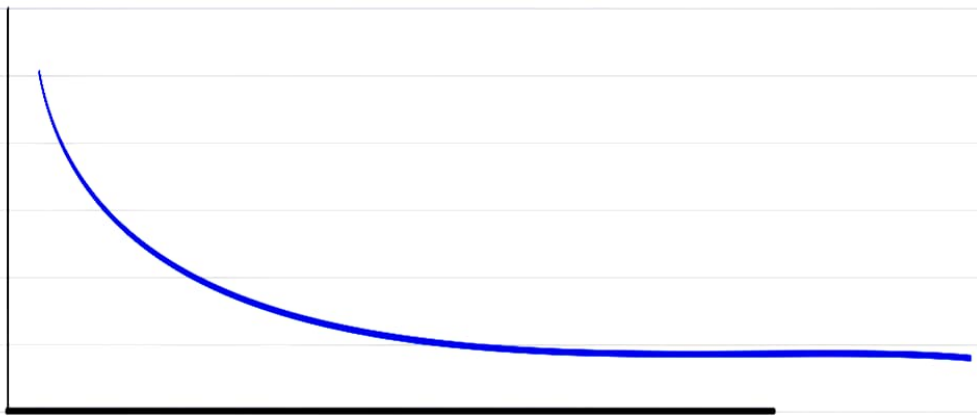


- The log-normal distribution describes the data whose logarithms are normally distributed.

- Log-transformation is a very popular method in ML for data preprocessing.

Pareto distribution

- its a continuous dist.
- entities having rare extreme events show pareto dist.
- it's heavily tailed distribution
- it has very high skewness



- it also shows significance of 80-20 rule i.e. 80% of information / value could be available in just 20% of data.